

Khulna University of Engineering & Technology
 Department of Building Engineering & Construction Management
 B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2015
Math 2123
 (Mathematics - III)

Full Marks: 210

Time: 3 hrs

- N.B.
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.

Section - A

1. (a) Define with example (any three): (12)
 (i) Equivalent matrix, (ii) Singular matrix, (iii) Trace, (iv) Symmetric matrix.
 (b) Point out three cases with example that matrices do not follow arithmetic (09)
 operations as real numbers.
 (c) Test linear dependence and independence of the following vectors. If possible (14)
 express the vectors as a linear combinations. Hence verify the linear
 combination (if exist); where, $[0, 1, 2, 3]$, $[2, 8, 9, 1]$, $[2, 6, 5, -5]$.
2. (a) Reduce the following matrix to echelon form, canonical form and normal form. (17)

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 4 & 2 & 0 & 2 \end{bmatrix}$$
 Hence find its rank.
 (b) Is the following matrices are invertible, why? $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $\begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$. (06)
 (c) If possible find the inverse of A. Hence express A as well as A^{-1} (if possible) as (12)
 a product of elementary matrices; where, $A = \begin{bmatrix} 0 & 1 \\ 2 & 4 \end{bmatrix}$.
3. (a) What is the value of λ so that the following system of linear equations has (i) (7)
 Unique solution, (ii) Many solution, (iii) No solution? where, $\lambda x + y = 2$
 $x + \lambda y = 2\lambda$
 (b) Find the largest Eigen value and corresponding Eigen vector of the Markov (18)
 matrix, $A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$. Hence verify. Also find the largest Eigen value of A^{10}
 and corresponding Eigen vectors (by using property).
4. (a) Write down three properties of Eigen value. (06)
 (b) Using Cayley-Hamilton theorem find the value of A^2 and A^{-2} (if possible) of the (09)
 matrix, $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$.
 (c) Define trivial and non-trivial solution of a set of homogeneous equations. Find (20)
 the non-trivial solutions of $4x_1 - x_2 + 2x_3 + x_4 = 0$, $2x_1 + 3x_2 - x_3 - 2x_4 = 0$,
 $7x_1 - 4x_2 - 5x_3 = 0$, $2x_1 - 11x_2 + 7x_3 + 8x_4 = 0$.

Section – B

5. (a) Find the velocity and acceleration of a particle which moves along the curve, $x = 2\sin 3t$, $y = 2\cos 3t$, $z = 8t$ at any time $t > 0$. Also find the magnitude of the velocity and acceleration along x axis at $t = \frac{\pi}{6}$.

(b) If $\frac{d^2 \vec{q}}{dt^2} = 12t\hat{i} + 24t^2\hat{j} + \sin t\hat{k}$ such that $\vec{q}(0) = \hat{i} + \hat{j}$ and $\frac{d\vec{q}}{dt} = 2\hat{i} + 3\hat{j}$, $t = 0$. (16)

(c) What are the physical meaning of ∇f and $\nabla \cdot \vec{F}$?

6. (a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$. (11)

(b) Show that $\vec{F} = (y^2 + 2xz^2)\hat{i} + (2xy - z)\hat{j} + (2x^2z - y + 2z)\hat{k}$ is irrotational and (13)
hence find its scalar potential.

(c) Calculate $\int_C \vec{F} \cdot d\vec{r}$, where C is the part of the spiral $\vec{r} = \langle a \cos \theta, a \sin \theta, a \theta \rangle$ corresponding to $0 \leq \theta \leq \frac{\pi}{2}$ and $\vec{F} = r^2 \hat{i}$.

7. (a) Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ over the surface of (12) $x^2 + y^2 + z^2 = 1$ which lies in the first octant.

(b) For Laplace transform, write following properties: First shifting, Second (10)
shifting, Scaling, Multiplication by powers of t , Division by t . Also state the
convolution theorem for inverse Laplace transform.

(c) Find the Laplace transform of $\frac{\sin at}{t}$. Does the transform of $\frac{\cos at}{t}$ exist? (13)
Justify.

8. (a) Find Laplace transforms of: (i) $t^2 \sin 2t$, (ii) $e^{-4t} \frac{\sin 3t}{t}$, (iii) Unit step function. (11)

(b) Find the inverse Laplace transforms of: (i) $\frac{s^2+3}{s(s^2+9)}$, (ii) $\frac{s}{s^2+4s+13}$, (iii) $\frac{e^{-s}}{s^2+1}$ (15)

$$\log \frac{s^2 - 1}{s^2}$$

(c) Using the convolution theorem, find $L^{-1}\left\{\frac{s^2}{(s^2+1)(s^2+2)}\right\}$

